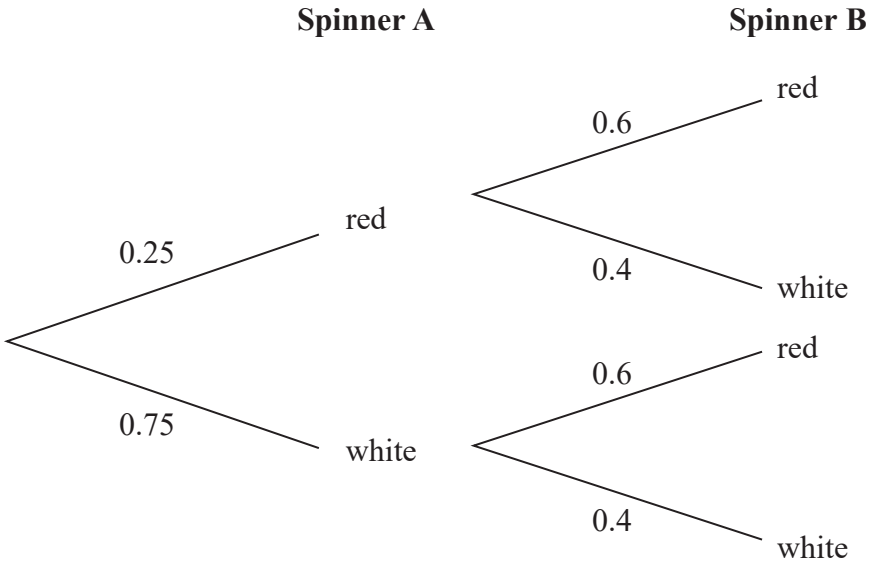
12 Alan has two spinners, spinner **A** and spinner **B**.
Each spinner can land on only red or white.

The probability that spinner **A** will land on red is 0.25
The probability that spinner **B** will land on red is 0.6

The probability tree diagram shows this information.



Alan spins spinner **A** once and he spins spinner **B** once.
He does this a number of times.

The number of times **both** spinners land on red is 24

Work out an estimate for the number of times **both** spinners land on white.

P(both red) = 0.25 × 0.6 = 0.15

1 mark
Multiply along the red–red path

P(both white) = 0.75 × 0.4 = 0.30

1 mark
Also multiply along the white–white path

White–white is twice as likely, so expect twice as many:

P	Times
0.15	24
0.3	48

×2 ×2

24 × 2 = 48

1 mark
Correct estimated frequency

48

(Total for Question 12 is 3 marks)